

ASafeSight: A PPE Detection System Using Deep Learning for Worksite Safety Compliance

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1. Problem Statement and Scope

Workplace safety is a critical concern in industrial environments such as construction sites, factories, and warehouses. Personal Protective Equipment (PPE), including hard hats, safety vests, and face masks, is required to reduce the risk of injuries and accidents. However, monitoring PPE compliance often relies on manual supervision, which can be inefficient and prone to human error, especially in large or busy work environments.

This project aims to develop an automated PPE detection system using computer vision and deep learning techniques. The system will analyze images from workplace environments and detect whether workers are wearing required protective equipment such as hard hats, masks, or safety vests. Object detection models will be used to locate and classify PPE items within images.

The scope of the project focuses on detecting PPE items in static images using object detection models. The system will not perform facial recognition or identify specific individuals. Instead, it will detect safety equipment within workplace scenes to assist in monitoring PPE compliance.

The effectiveness of the model will be evaluated using several metrics commonly used in object detection tasks. These include mean Average Precision (mAP@0.5), precision-recall curves, and detection latency. The mAP metric measures how accurately the model detects and localizes objects across different classes. Precision-recall curves will provide insights into detection performance across confidence thresholds, while latency will measure inference speed.

Several constraints must also be considered. Training time must remain within the project's computational limitations, and the model should be efficient enough to run within reasonable inference times. The dataset size may also be limited, which could affect generalization performance. Therefore, transfer learning using pretrained detection models will be used to improve training efficiency.

2. Related Work

Workplace safety incidents remain a critical concern in industrial and construction sectors, where many injuries result from non-compliance with Personal Protective Equipment (PPE) standards. Traditional manual monitoring methods are often labor-intensive, inconsistent, and prone to human error, especially in large-scale environments. These limitations have led to the development of automated computer vision systems that utilize deep learning for real-time safety monitoring. Studies have shown

that Convolutional Neural Networks (CNNs) are highly effective in detecting PPE such as helmets, safety vests, and masks in complex environments [3][6].

Among the various deep learning models, the YOLO (You Only Look Once) architecture has emerged as a leading framework for real-time object detection due to its balance of speed and accuracy. Earlier research focused on customized YOLOv8 models to improve detection performance in construction site scenarios [1][4]. More recent studies have explored advanced versions such as YOLOv11, which offers improved inference speed and is well-suited for edge deployment environments [2]. These advancements highlight the continuous evolution of object detection models toward higher efficiency and real-time applicability.

To further enhance detection performance, researchers have introduced architectural improvements. For instance, integrating advanced backbones and feature extraction techniques improves robustness in detecting partially occluded objects. Additionally, modern approaches incorporate multi-scale processing and attention mechanisms to address challenges such as varying lighting conditions and geometric distortions [6]. These innovations contribute to more reliable PPE detection in real-world industrial settings.

Beyond detection, Reinforcement Learning (RL) has been explored to improve system adaptability. RL-based approaches enable dynamic thresholding, allowing the system to adjust detection sensitivity based on environmental conditions, reducing false positives and improving anomaly detection [5]. Furthermore, RL techniques have been applied to enhance low-quality images, ensuring consistent detection performance even in poor visibility or motion blur conditions [7].

In addition, Natural Language Processing (NLP) plays a role in transforming detection outputs into actionable insights. Previous studies have utilized NLP to analyze safety reports and extract meaningful patterns from textual data, supporting decision-making in safety management [3].

Overall, these studies demonstrate that while CNN-based models such as YOLO provide strong capabilities in visual detection, the integration of adaptive technologies like RL and NLP is essential for developing a comprehensive and intelligent safety system. Building on these advancements, the proposed ASafeSight system aims to combine a fine-tuned YOLOv8 model with RL-based adaptive thresholding and an NLP-driven safety auditing component to enhance PPE compliance monitoring in industrial environments.

3. Dataset Plan

The primary dataset is sourced from the Construction Site Safety (CSS) dataset available via Roboflow. Licensed CC BY 4.0 (Attribution required, allows commercial and academic use).

Consent: Since the dataset consists of public domain images and industry-standard safety simulations, individual consent is handled by the original dataset aggregators. ASafeSight further anonymizes data by discarding raw video after processing.

Representativeness: The dataset includes diverse lighting conditions, various PPE colors, and different worker postures to ensure the model generalizes across real-world construction sites.

4. Planned CNN/NLP/RL Components and MVP

CNN (Computer Vision): Implementation of YOLOv8 for real-time multi-class object detection (Hardhat, Vest, Mask, No-PPE).

NLP (Safety Auditor): A BERT-based classifier used to process textual safety logs and map them to specific visual violation detections.

RL (Adaptive Thresholding): A Q-Learning or Policy Gradient agent that adjusts the confidence thresholds of the CNN dynamically based on ambient lighting and environment complexity.

MVP (Minimum Viable Product): A functional pipeline that accepts a video feed, detects PPE violations, and generates a timestamped text summary of the event.

5. Ethics Risk Register

The researchers has identified several potential risks and corresponding mitigation strategies to ensure safe and ethical deployment. One potential ethical concern is privacy. Workplace monitoring systems that analyze images may raise concerns about employee privacy. To mitigate this, the system will focus solely on detecting protective equipment rather than identifying individuals. No facial recognition or personal identification will be implemented. Additionally, constant surveillance will process data in real-time only, with no storage of raw facial data, and facial blurring will be applied post-detection. Another risk involves false positives or false negatives if the model misclassifies objects or fails to detect PPE items, potentially leading to incorrect safety assessments.

To reduce this risk, the system will report detection confidence scores and emphasize that the system serves as a supportive monitoring tool rather than a replacement for human supervision. Dataset bias may occur if training data is limited to certain workplace environments, causing the model to perform poorly in different conditions. To address this, the team will audit datasets for demographic and environmental diversity, apply data augmentation techniques, and evaluate model performance across varying image conditions. Manual overrides will also be provided for low-confidence detections to further reduce bias. Finally, safety logs could potentially be leaked, and mitigation measures include encrypted storage for all

violation logs and strict access controls to ensure data confidentiality.

6. Team Roles & Milestones

The team comprises three members from Holy Angel University's School of Computing, enrolled in Intelsys. Roles are assigned to cover the full project lifecycle

1. **Paulo Miguel R. Limpin** – Project Lead / Integration
 - Scope definition, sprint planning, system integration, Model Card authoring, and defense coordination
2. **Gabriel Magat Jr.** – Data & Ethics Lead
 - Dataset sourcing and licensing, preprocessing pipelines, bias checks, ethics statement, GitHub Creator
3. **Charles Pascua** – Modeling & Evaluation Lead
 - YOLOv8 fine-tuning, NLP module design, CNN ablations, mAP/PR reporting, latency benchmarks, RL agent evaluation

Milestones

- **Week 1:** Proposal & GitHub setup (v0.1)

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